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Gruppo Nazionale Struttura della Materia, C.N.R., Milano***Intrinsic Optical Properties of CrCl_3 ¹⁾**

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Dedicated to Prof. Dr. Dr. h.c. P. GÖRLICH, on the occasion of his 65th birthday

The intrinsic optical absorption spectrum of CrCl_3 has been investigated in the 0.6 to 6 eV region at temperatures from 80 to 300 °K. A careful analysis of the crystal field bands in the 0.6 to 3 eV region has revealed some structures and details, such as an Urbach's rule edge, not previously reported. The optical anisotropy of CrCl_3 was also studied over the same spectral range. In the region between 3 and 8 eV reflectivity spectra were also obtained. The refractive index of CrCl_3 has been determined on evaporated films from 1000 to 380 nm [$n_{(500\text{ nm})} = 1.85 \pm 0.1$]. The spectral distribution of photocurrent at $\approx 80^\circ\text{K}$ was measured and showed weak peaks associated with the main crystal field bands and a sharp increase at shorter wavelengths; maximum photoconductivity yield was reached at (3.2 ± 0.2) eV. The data on the crystal field bands are discussed and some new structures are assigned to specific electronic transitions. Account is taken of the spin-orbit interaction among the various multiplets and the trigonal distortion of the octahedral symmetry of the Cr^{3+} environment. A discussion of the optical and photoconductivity data in the 3 to 8 eV region is given and alternative explanations of the absorption edge at 3.2 eV are proposed.

Le spectre d'absorption optique de CrCl_3 a été étudié dans la région spectrale 0,6 à 6 eV, entre 300 et 80 °K. Les bandes d'absorption dans la région d'énergie 0,6 à 3 eV montrent des structures et détails (règle d'Urbach) jamais observés auparavant. L'anisotropie optique de CrCl_3 a été étudiée dans le même intervalle d'énergie, tandis que le spectre de réflexion a été mesuré jusqu'à 8 eV. Le spectre de photoconductivité à environ 80 °K montre de faibles maxima associés aux bandes principales de la région visible du spectre et une forte absorption aux plus courtes longueurs d'onde; le maximum de photocourant a été trouvé à $(3,2 \pm 0,2)$ eV. Les résultats obtenus dans la région spectrale 0,6 à 3 eV ont été discutés, considérant aussi le couplage spin-orbite et la distortion trigonale de l'entourage de l'ion Cr^{3+} . Une discussion des données optiques et photoélectriques entre 3 et 8 eV a été faite et des explications alternatives pour le bord d'absorption à 3,2 eV ont été suggérées.

1. Introduction

Chromium trihalides have recently received a good deal of attention because of their peculiar physical properties and possible applications [1 to 5]. These materials are insulators and become ferromagnetic below their Curie temperature. They have a typical layer structure [1] and unfilled d-shells, the presence of which determines the magnetic properties. Until recently [6] the studies of the intrinsic optical properties were limited to the so-called "crystal field

¹⁾ A preliminary report of this work has been given at the "International Symposium on Anisotropy in Layer Structures" Taormina, Italy, September 1968.

bands", produced by the d-d dipole-forbidden intraconfigurational transitions of the chromium ion [5]. Various sharp peaks present in the paramagnetic [5] and ferromagnetic [1] phases and the main broad bands were attributed to the spin-forbidden quartet-to-doublet transitions and to the spin allowed quartet-to-quartet transitions, respectively. In particular the latter bands were assigned to transitions ${}^4A_2 \rightarrow {}^4T_2$ and ${}^4A_2 \rightarrow {}^4T_1$ in the order of increasing energy [5].

After a detailed study of the magneto-optical properties of CrBr_3 in the "crystal field" region and in the steep absorption edge at higher energies, Dillon et al. [1] attributed the absorption edge to a "charge transfer" transition or, in other words, to a jump of the electron from the Cl p-orbitals to a Cr d-level. In recent reflectivity studies on CrBr_3 and CrI_3 , Grant and Street [6] reached similar conclusions as to the transitions responsible for the absorption in the same region of the spectrum.

In the present work we show our optical and photoconductivity data on CrCl_3 . In particular the photoconductivity data appear to be the first reported for any transition metal halide. Our principal aim is to gain information to help assess the relevancy of the molecular orbital and energy band theories in different parts of the spectrum.

Section 2 is devoted to a description of the various techniques used. In Section 3 our results are shown, and in Section 4 our new measurements on the crystal field bands and on the strong absorption region at higher energies are discussed.

2. Experimental

2.1 Absorption and reflectivity

All the visible and near ultraviolet (up to 6 eV) absorption spectra have been measured on a double beam recording spectro-photometer (type Cary 14). The specimens were mounted inside a stainless steel cryostat. Polarized absorption data were obtained by inserting a properly mounted Rochon prism in the light path.

Absorption coefficients of the order of 10^4 to 10^5 cm^{-1} were measured for single crystals. A stripping technique in which self-adhesive tape was used to cleave the crystals provided us with samples of different thickness down to $\approx 1 \mu\text{m}$. The crystal was floated off its adhesive backing by immersion in chloroform and transferred onto a quartz support.

Reflectivity data over the energy range 0.8 to 6 eV were obtained using a Cary 14 spectro-photometer, modified to work in reflection. Measurements were made using unpolarized light incident on the specimen surface at an angle of 25° with respect to the *c*-axis of the crystal. Data beyond 6 eV were obtained with a vacuum uv Hilger grating monochromator; in this case the angle of incidence was 12.5° .

Films were also grown inside the cryostat, and their thickness was continuously monitored by means of interference fringes obtained by reflection.

The refractive index of the films, *n*, was measured by the well known interference fringe technique which involves the determination of *n* at a fixed λ_0 together with the film thickness. The main source of error with this method is the fact that the reflectivity of a film may be lower than that of a single crystal, as thin films are somewhat porous.

2.2 Photoconductivity

Photoconductivity data were obtained by using a Cary 31 vibrating reed electrometer and by employing the "rate of charge" method. To make electrical contacts with the crystals, we interposed indium strips between two phosphor bronze springs and the crystal surface. The separation of the electrodes in a arrangement of the "surface cell" type was 1 to 2 mm. The entire area of the crystal between them ($\approx 5 \text{ mm}^2$) was illuminated by light from a monochromator. The exciting light was incident onto the crystal surface in a direction parallel to the c -axis. All the measurements reported below were carried out with current flowing perpendicular to the c -axis of the crystal, i.e. in the plane of the layers. This was done because the electrical conductivity in a direction perpendicular to the layers cannot be described with the standard concept of carrier mobilities. The electric field used was of the order of 1000 V/cm . The data were corrected for light intensity, the calibration being made with the aid of a copper-constantan Eppley thermopile.

3. Results

3.1 Optical measurements in the crystal field region

Fig. 1 shows the dispersion of the average refractive index at room temperature. The reason why we call it average is due to the fact that the microcrystals forming the film have their hexagonal c -axis randomly oriented with respect to the incident light. In the figure, we notice two small humps related to the two main absorption bands in the visible region of the spectrum whose peak energies at 300°K are indicated by arrows. The dashed line represents the values of n if the two crystal field bands did not exist. With an approximate Kramers-Kronig calculation and supposing that the crystal field bands were Gaussian in shape, we checked, a posteriori, that the order of magnitude of the effect induced by the absorption bands on the refractive index was correct.

Fig. 2 shows a composite absorption curve of CrCl_3 in the crystal field region of the spectrum. Samples of various thicknesses were used to obtain such a spectrum. Data were not corrected for reflectivity. Details on the nomenclature we used for the various peaks seen in Fig. 2 will be given in the next section. Here we anticipate that our choice comes from considering the "severe trigonal distortion" [1] of the octahedron of Cl ions surrounding Cr^{3+} . ${}^4\text{T}_2$ and

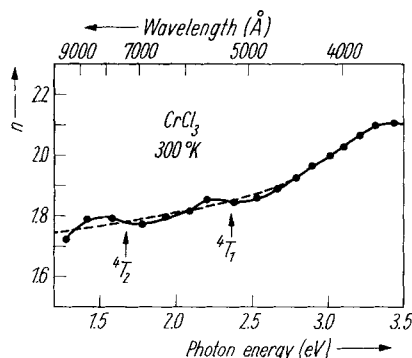


Fig. 1. The average refractive index of a CrCl_3 film plotted against photon energy at room temperature. The positions of the main visible bands are also indicated

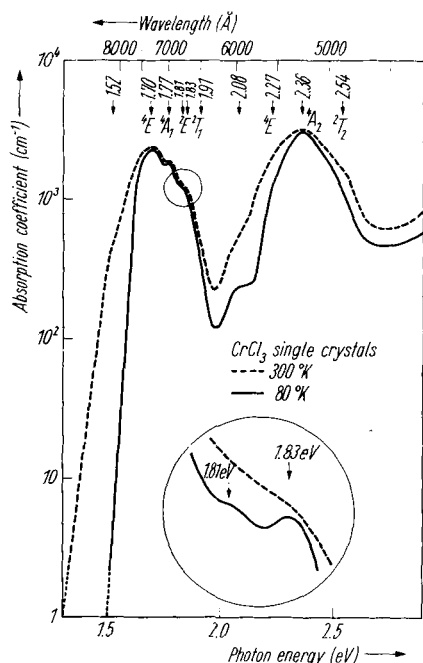


Fig. 2. The variation of absorption coefficient with photon energy for CrCl_3 crystals at 300 and 80 °K. All the structures are indicated by arrows; the values refer to the 80 °K curve; the new assignments are also reported

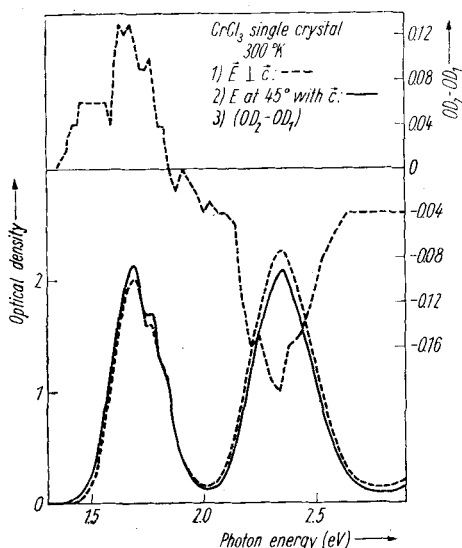


Fig. 3. The optical density of a single crystal of CrCl_3 at room temperature for $E \perp c$ and E at 45° with c . The direction of light propagation always makes an angle of 45° with the c -axis. Also shown is the difference $(OD_2 - OD_1)$ calculated from the original spectrophotometer traces

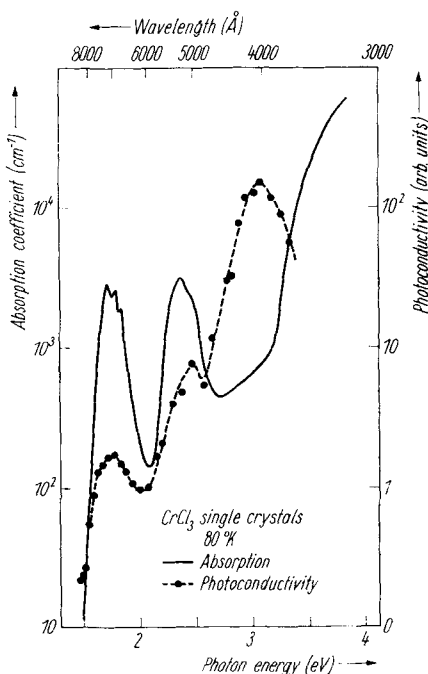
4T_1 are the cubic field contributions to the main crystal field bands (if the perturbing trigonal field is not considered). The new features not previously reported [1, 5] include: (i) the energy dependence of the first edge following the Urbach's rule [7]; (ii) new structures superimposed on the ${}^4A_2 \rightarrow {}^4T_2$ band (particularly evident in the spectrum of the thinnest crystals taken at 80 °K) at energies of 1.52, 1.81, 1.91 eV, and on the ${}^4A_2 \rightarrow {}^4T_1$ band at 2.08 and 2.27 eV. Furthermore we observe a temperature dependence of the oscillator strength almost solely due to line broadening effects. A small temperature shift of the absorption peak energy exists, but it is not sufficient to be visible in Fig. 2. We measured for the 4E peak a value of 1.68 eV at 300 °K, and 1.70 eV at 80 °K; for the 4A_2 peak 2.35 eV at 300 °K, and 2.36 eV at 80 °K.

In Fig. 3 we show our linearly polarized light absorption measurements in which E , the electric vector of the incident radiation, made first 90° with the c -axis and then 45° in order to have a component parallel to c ; the direction of light propagation was always 45° with respect to c . CrCl_3 flakes are not perfectly flat and hence mainly qualitative aspects had to be considered. The most noticeable effect is the optical anisotropy of the crystal field bands. The lower energy band is more easily excited by a quantum with E parallel to c . Quantitatively the effect is certainly greater than shown in Fig. 3 (at least twice), because the absorption relative to the curve 2 of Fig. 3 has a component $E \perp c$.

Fig. 4. The spectral distribution of photocurrent in a single crystal of CrCl_3 at liquid air temperature, together with absorption coefficient for comparison

3.2 Photoconductivity measurements

In Fig. 4 our 80 °K photoconductivity data are shown (together with the absorption spectrum) versus photon energy. We can see that for a similar value of μ , say 10^3 cm^{-1} , photoconductivity is one hundred times higher at $\approx 3.2 \text{ eV}$ than at $\approx 1.6 \text{ eV}$. The large maximum of photocurrent occurs at about the same energy as the absorption edge located at $3.2 (\pm 0.2) \text{ eV}$. We can now comment on the decrease of the photoconductivity at energies greater than 3.1 eV and suggest, as is currently done in the literature, that the increase in absorption coefficient produces a strong absorption at the surface where a smaller "schubweg" is likely to occur.



3.3 Optical measurements in the 3 to 8 eV region

To study the absorption spectrum in regions of high absorption coefficient, we prepared thin films by evaporation onto substrates at room temperature. In Fig. 5 we show the absorption curves of our "thick" and "thin" films. The crystal field bands appear to be rather structureless, probably due to the disordered structure of the films. Two bands located at ≈ 4.1 and $\approx 4.9 \text{ eV}$ are found and their intensity is more than one order of magnitude greater than that of the crystal field bands.

Reflectivity spectra were also obtained and are shown in Fig. 6 (we recall that data were taken with the direction of the incident light as near to the c -axis as possible). The reflectivity is given in arbitrary units since the specimens are not flat and the surfaces of different crystal areas are at different angles with respect to the incident light. Besides, it is unavoidable that, upon lowering the temperature, the stresses acting on CrCl_3 flakes change the inclination of these various sub-surfaces, and thus making a quantitative comparison among reflected intensities at different temperatures unrealistic. Notwithstanding these drawbacks, we believe that our reflectivity results are essentially correct since they show the same general features (apart from an almost rigid shift to higher energies) as CrBr_3 and CrI_3 [6] crystals.

In Fig. 6a the solid and dashed lines represent the data obtained after a correction has been made for the contribution of the light reflected from the second surface of the specimen. This correction is particularly relevant in the regions where the absorption coefficient is lower [8]. The dotted lines, however, show the actual recording taken from the spectro-photometer. Reflectivity

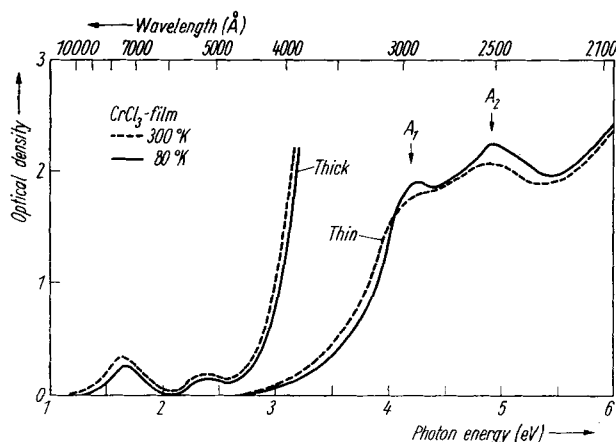


Fig. 5. Optical absorption spectra for thick and thin films of CrCl_3 deposited on SiO_2 substrates at room temperature. Full and dashed lines indicate spectra measured at 80 and 300 °K, respectively

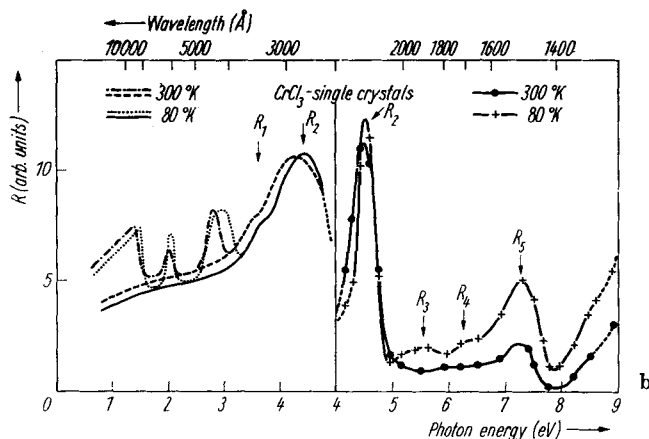


Fig. 6. The reflection spectra of single crystals of CrCl_3 at 80 and 300 °K. The curves of a) and b) were obtained on two different experimental sets

peaks are denoted by R_i , ($i = 1, 2, \dots, 5$). A knee at ≈ 3.8 eV and a peak at ≈ 4.5 eV appear to correspond to the structures seen in absorption (see Fig. 5) in the 4 to 5 eV region.

4. Comments and Discussion

4.1 The crystal field band region

The new data we obtained in the crystal field region are summarized as follows:

(i) at the first absorption edge (≈ 1.5 eV at 80 °K), the variation of the absorption coefficient with photon energy is given by Urbach's rule [7];

(ii) in the main bands, for which we confirm the absolute value of μ and the general shape obtained in [5], new structures were found. The oscillator strength is noticeably temperature dependent. Whereas, the changes of the absorption peak energies with temperature are very small;

- (iii) the polarized absorption spectra showed optical anisotropy in the two crystal field bands;
- (iv) the dispersion of the average refractive index from 1.5 to 3.5 eV at room temperature was obtained;
- (v) a weak photoconductivity was excited in the crystal field bands.

The fact that we find an Urbach's rule behaviour at the first edge of CrCl_3 is a further confirmation of the general character of this rule, which occurs in such different situations as the intrinsic absorption edge in CdS and alkali halides, bound excitons in alkali halides, anthracene, and impurity states [9]. The presence of an Urbach's rule at the low energy side of the ${}^4\text{T}_2$ band is of considerable importance in our case since it should exclude the fact that the structure in the ${}^4\text{T}_2$ may be due to phonons. The various Urbach's rule explanations imply essentially a multi-phonon process [7, 10] and, therefore, absence of any vibrational structure. Further, if we choose an heuristic attitude, the exponential behaviour on the low energy side of the absorption edge increases the probability of observing the above mentioned structure, the absence of which has been demonstrated. To complete the argument we add that in general [11] the vibrational structures in solids are highly temperature dependent, but in this case no strong temperature variation was found.

We now proceed to consider the two bands with their superimposed structures. The Laporte's rule, forbidding transitions between levels of the same parity (in this case d-d levels), is invalidated because in this compound the chromium site is not an inversion centre. Most of the intensity of the band is then due to electric dipole transitions allowed by the mixing of odd parity states through the static crystalline field. Lattice vibrations will then broaden the bands and, through dynamical mixing, add some extra oscillator strength to the transitions. Such a contribution is rather small, relative to the one which lattice vibrations are able to give to forbidden transitions allowed essentially by the dynamical mixing. The strong temperature dependence of the oscillator strength [12] is typical of the second situation.

We move now to detailed analysis of the spectrum of Fig. 2. If we consider the Sugano and Tanabe diagrams [13] for d^3 electrons with $10 Dq = 13,700 \text{ cm}^{-1}$ the order in which the various bands are found is rather well fixed: ${}^4\text{T}_2$, ${}^2\text{E}$, ${}^2\text{T}_1$, ${}^4\text{T}_1$, ${}^2\text{T}_2$. Furthermore we know that in CrCl_3 the octahedron of Cl ions around Cr^{3+} has a severe trigonal distortion [1] which, together with the spin-orbit couplings among various states, has the effect of removing a certain amount of the degeneracy. Finally, through careful absorption measurements in the crystal field band region, we have found new structures. Some of these new structures could be assigned to certain electronic transitions taking into account that a trigonal field splits the ${}^4\text{T}_2$ levels in ${}^4\text{A}_1$ and ${}^4\text{E}$ and the level ${}^4\text{T}_1$ in ${}^4\text{A}_2$ and ${}^4\text{E}$ [14, 15] (for these levels the spin-orbit interaction has a very small influence on the splitting). The peaks at 1.70 and 1.77 eV are then assigned to the levels ${}^4\text{A}_1$ and ${}^4\text{E}$, respectively. We further know [14] that the combined effects of the spin-orbit interaction and the trigonal field perturbation produce a splitting of the ${}^2\text{E}$ level. Since we have experimentally found a structure at 1.81 eV, we propose that the 1.81 and 1.83 eV bands are due to the split ${}^2\text{E}$ level. We realize that a value of $\approx 160 \text{ cm}^{-1}$ for the splitting of the ${}^2\text{E}$ level is too large but if one takes into account the considerable hybridization of ${}^4\text{T}_2$

and 4T_1 with 2E (Wood et al. [5] propose 16%) and makes detailed calculations following Sugano and Tanabe [14], better agreement may be reached.

The band at ≈ 1.91 eV should be due to 2T_1 spin forbidden quartet-to-doublet transition and be split in the same way as the 2E band. However, both bands and, even more so, the splittings are observable with difficulty because of the low oscillator strength for this transition.

On the 1.52 and 2.08 peaks we do not have definite suggestions, for the moment.

We propose furthermore, that the 2.27, 2.36, and 2.54 peaks correspond to transitions to the 4E , 4A_2 , and 2T_2 levels, respectively. As far as the assignments of 2.27 and 2.36 peaks are concerned, we have chosen the sequence 4E , 4A_2 , instead of its reverse, because the CrCl_3 optical anisotropy has the same sign as the one of emerald [16] and opposite to that of ruby [14, 15]. Since the sign depends on the order of 4E and 4A_2 levels, we feel justified in attributing these levels in this way. The explanation of the opposite signs of the emerald and ruby optical anisotropies has also been given because of the importance of the A_2^0 static field parameter in determining the value and sign of the parameters v and v' [16, 17]. From our data we find that the difference ${}^4A_1 - {}^4E \approx 600 \text{ cm}^{-1}$ and ${}^4E - {}^4A_2 \approx 700 \text{ cm}^{-1}$, and, following Macfarlane [15], we easily calculate that $v \approx -1200 \text{ cm}^{-1}$ and $v' = +1300 \text{ cm}^{-1}$.

Both the optical anisotropy and the splittings appear to be qualitatively accounted for. Qualitative relations between v and v' and the experimentally measured optical anisotropy are still lacking.

With regard to the small photoconductivity peaks, corresponding to the crystal field bands, we think that after excitation a hopping type of conduction takes place in which the electron jumps between nearby Cr ions. This type of process will be discussed further in the following.

4.2 The high absorption coefficient region

Let us consider the photoconductivity data. First it is necessary to explain why photons with energy ≈ 3.2 eV are two orders of magnitude more efficient in producing photocurrents than the 1.6 eV photons. If the two types of quanta are absorbed with the same probability (see Fig. 4), and therefore excite an equal number of electrons, a careful study of the mobility (and therefore also of the type of transport process) should be made in order to understand the wavelength dependence of photoconductivity previously outlined. However, if it is assumed that the edge at ≈ 3.2 eV is due to the charge transfer transition $3p(\text{Cl}) \rightarrow 3d(\text{Cr})$, two situations might occur:

a) the electron on jumping to a Cr ion leaves behind a trapped hole localized at a Cl ion, and so forms a Cr^{2+} ion. Transport is then achieved by means of thermally activated hopping of the electron from a Cr^{2+} to a Cr^{3+} ion and so on. The following reactions show the way this could happen:

- (i) $\text{Cl}(3p^6) + \text{Cr}^{3+}(3d^3) + h\nu \rightarrow \text{Cl}(3p^5) + \text{Cr}^{2+}(3d^4)$,
- (ii) $\text{Cl}(3p^5) + \text{Cr}^{2+}(3d^4) + kT \rightarrow \text{Cl}(3p^6) + \text{Cr}^{3+}(3d^3) + e^-$.

This would be quite analogous to what is observed for $\alpha\text{-Fe}_2\text{O}_3$ and NiO [18]. The hole remains trapped and the electron now has the same mobility as one excited with a "crystal field" transition: for the same absorption coefficient and the same number of excitations the same value of the photoconductivity

should be observed. This process does not appear to explain the situation at energies in the region of 3.2 eV. Even if the hole underwent a hopping type transport, we would not obtain a much higher photocurrent.

b) The electron, after excitation, moves as explained in section a) but the hole, through thermal activation, becomes delocalized and free to move inside the valence band. In this case the photocurrent may be a hundred times higher than that in section a) and thus the high value of photoconductivity excited in this energy region would be justified. Two consequences of this process are as follows: a strong temperature dependence of the photoconductivity and a positive Hall coefficient.

Our photoconductivity data, therefore, appear to be rather well accounted for in terms of a charge transfer transition. A plausible explanation for the absorption edge at ≈ 3.2 eV has thus also been found.

We think, however, that other possible explanations, both of the absorption and of the photoconductivity data should not be disregarded, and so we offer an alternative proposal which is of a speculative nature for the moment. If the transition responsible for the 3.2 eV edge is a $3p(\text{Cl}) \rightarrow 4s(\text{Cr})$, both the conduction band electrons and the valence band holes can be involved in the transport process, and the high photoconductivity will be clearly and naturally justified. The main objection to this model would be that the $3p(\text{Cl}) \rightarrow 3d(\text{Cr})$ transitions should have a lower energy [1] than the $3p(\text{Cl}) \rightarrow 4s(\text{Cr})$, and in the spectra of Fig. 2 and 5 no band would remain to be attributed to this process. We should now draw attention to the fact that for CrCl_3 a molecular orbital picture of the type proposed in [1] should be used with caution and particularly in the case of the $3p(\text{Cl})$ and $4s(\text{Cr})$ electrons a Bloch wave description would appear more appropriate. For certain values of the wave vector k in the Brillouin zone, there is no reason why the $3p(\text{Cl}) \rightarrow 4s(\text{Cr})$ transition should not involve a smaller energy than the $3p(\text{Cl}) \rightarrow 3d(\text{Cr})$ transition. In fact s and d type conduction bands are in opposite order from the point of view of energy at the point Γ and X in the Brillouin zone of the alkali halides [19]. To conclude the argument, we suggest that the charge transfer transitions might be hidden under the band to band transitions $3p(\text{Cl}) \rightarrow 4s(\text{Cr})$ and this might be possible since in certain cases (i.e. beryl doped with Cr^{3+} , K_2NaCrF_6 [5], and Cr^{3+} in corundum [13]) the charge transfer bands have an intensity comparable with the crystal field bands. In any case $3p(\text{Cl})$ wave functions have a stronger overlap (and hence a greater oscillator strength) with $4s(\text{Cr})$ than with $3d$ wave functions.

In order to make a choice between the two alternative proposals that we have just made on the 3 to 4 eV region of the absorption spectrum, we believe that more data, particularly on the transport properties of photoexcited carriers, are necessary.

Let us now consider the higher energy optical data. Absorption and reflectivity spectra show several peaks beyond the 3.2 eV edge, but at the moment we are not able to make definite correlations of the various peaks with specific charge transfer or interband transitions. This would, in fact, require a band structure calculation at least at points of high symmetry in the Brillouin zone. However, if we refer to the transitions between the orbitals of a chromium trihalide complex, as done by Grant and Street [6], we may attribute the first two reflectivity peaks to electron transfer transitions. Band to band ($3p(\text{Cl}) \rightarrow 4s(\text{Cr})$) transitions may then be responsible for the structure at higher energies.

These explanations will obviously be completely different if the tentative proposal we have put forward to explain the photoconductivity data would turn out to be correct.

Finally we may comment on the strong decrease in reflectivity at ≈ 4.8 eV which has also been found (shifted in energy) in CrBr_3 and CrI_3 [6]. In GaS and GaSe [20] a similar behaviour in the reflectivity spectrum was attributed to the absence of critical points in the joint density of states in the corresponding energy region.

Since the similarity between R and ϵ_2 and because

$$\epsilon_2(\omega) = \frac{n(\omega) c \mu(\omega)}{\omega},$$

where $n(\omega)$ is the real part of the refractive index, c the light velocity, and $\mu(\omega)$ the absorption coefficient, we tentatively attribute the reflectivity decrease to the influence of $n(\omega)$. This is suggested because in the energy region around 4.8 eV $\mu(\omega)$ (see Fig. 5 and 6) is increasing.

5. Summary

New structures found in the absorption spectrum over the crystal field region together with optical anisotropy measurements, allowed us to propose different and more detailed assignments of the optical transitions involved. The photoconductivity induced in the same energy region may be due to electrons hopping between nearby Cr ions.

With reference to the photoconductivity data at ≈ 3.2 eV, possible explanations of the absorption edge in that energy region have been discussed. By means of absorption and reflectivity data several peaks in the 4 to 8 eV region have been observed.

In order to clarify the role of the d levels in the band picture, and to obtain soundly based attributions of the peaks seen in the absorption and reflectivity spectra, more transport and optical property measurements on isostructural compounds are required.

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